4MA1_1F_ch3_sequences_functions_graphs

Nguyen Khac Kiem PhD

1. 4MA1/1F_Winter_2025_Q17

2. 4MA1/1F_Winter_2025_Q11

3. 4MA1/1FR_Summer_2025_Q11

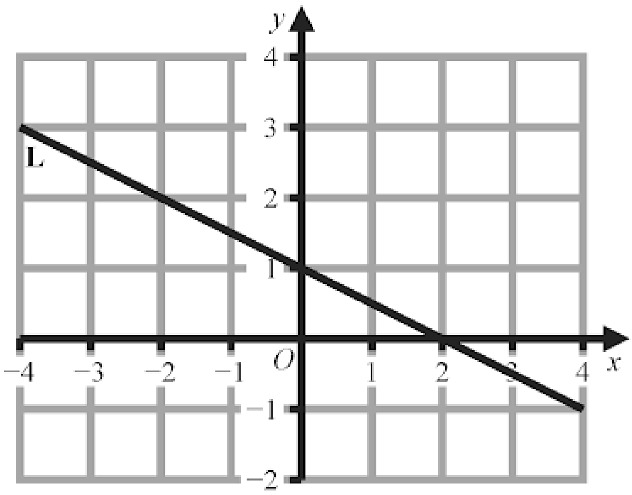
4. 4MA1/1FR_Summer_2025_Q9

5. 4MA1/1F_Summer_2025_Q8

6. 4MA1/1F_Summer_2025_Q6

7. 4MA1/1FR_Summer_2024_Q24

Line **L** is drawn on the grid.

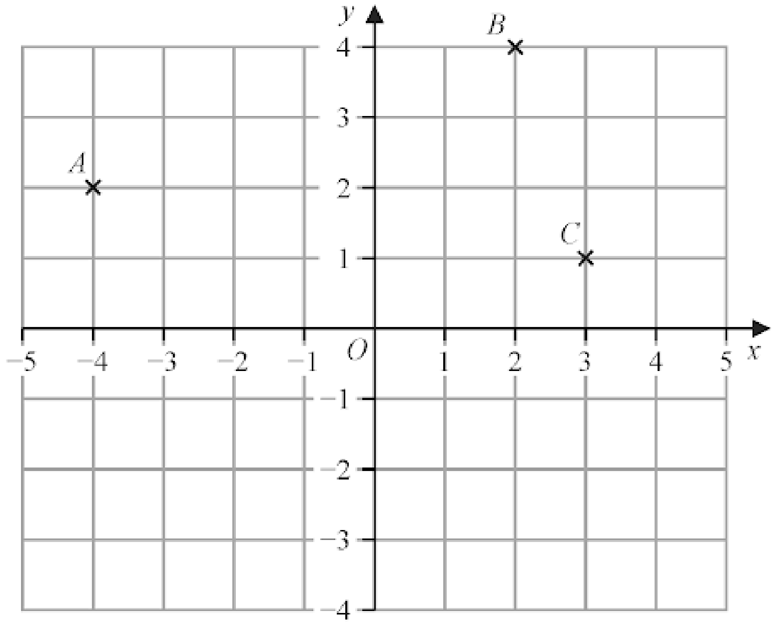


Find an equation for **L**
Give your answer in the form $y = mx + c$

(3)

8. 4MA1/1FR_Summer_2024_Q9

The diagram shows three points, A, B and C, marked on a grid.



(a) Write down the coordinates of A

(.................... , ...)
(1)

(b) Find the coordinates of the midpoint of AB

(.................... , ...)
(2)

D is the point on the grid so that $ABCD$ is a rectangle.

(c) Find the coordinates of D

(.................... , ...)
(2)

9. 4MA1/1F_Summer_2024_Q16

Here are the first four terms of an arithmetic sequence.

$$1 \qquad 4 \qquad 7 \qquad 10$$

(a) Find an expression, in terms of n, for the nth term of this sequence.

(2)

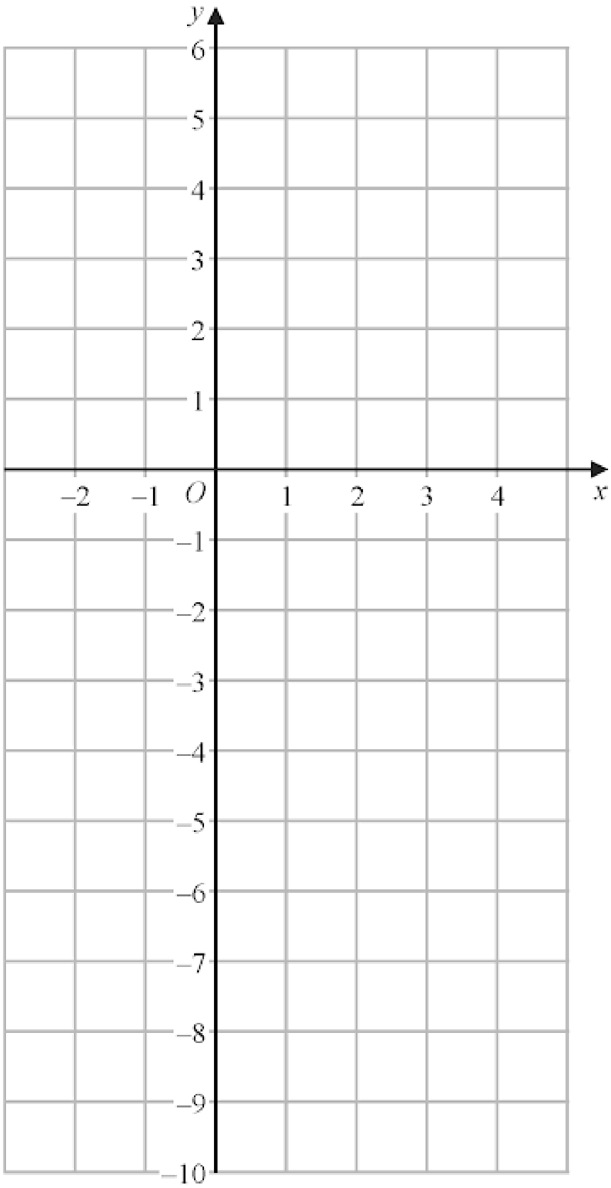
The nth term of a different arithmetic sequence is $5n + 17$

(b) Find the 12th term of this sequence.

(1)

10. 4MA1/1F_Summer_2024_Q13

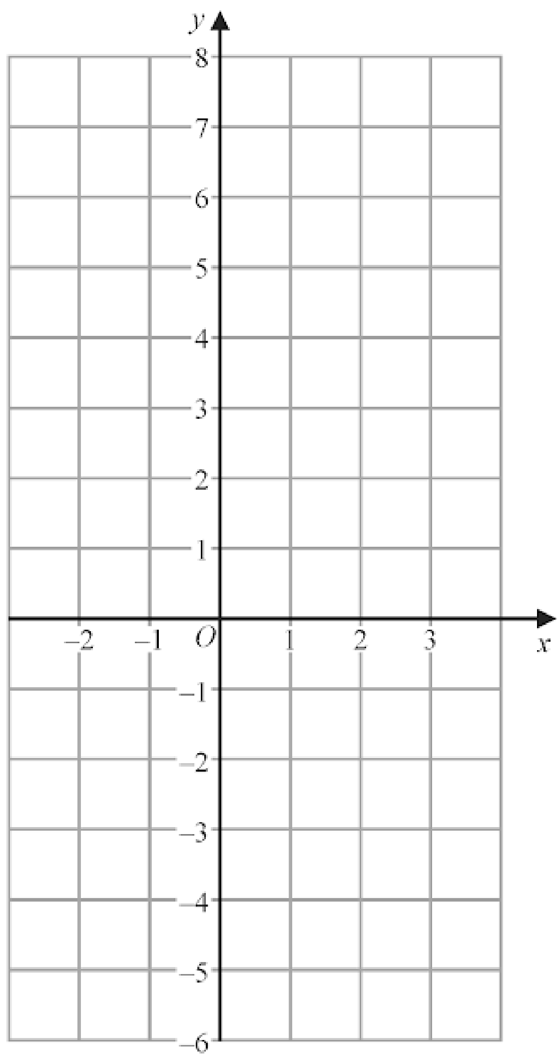
On the grid, draw the graph of $y = 2x - 3$ for values of x from -2 to 4



(3)

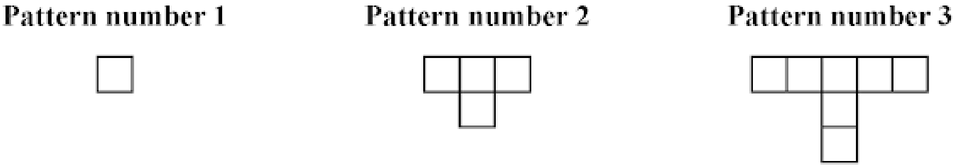
11. 4MA1/1F_Winter_2023_Q11

On the grid below, draw the graph of $y = 1 - 2x$ for values of x from -2 to 3



3 marks

A sequence of patterns is made from squares.



(a) In the space below, draw Pattern number 4

Pattern number 4

(1)

(b) Complete the table.

Pattern number	1	2	3	4	5
Number of squares	1	4	7		

(1)

(c) Work out the number of squares in Pattern number 8

....

(1)

Angus says

"there are 42 squares in Pattern number 15"

Angus is incorrect.

(d) Explain why.

....

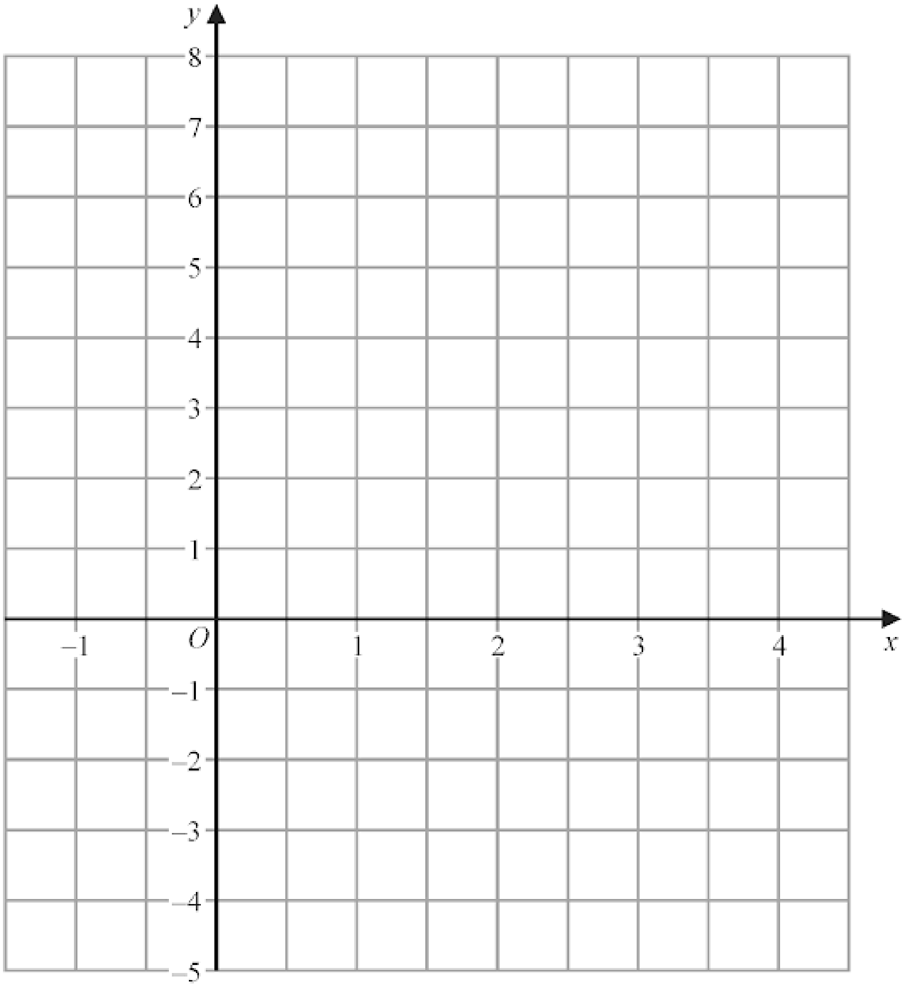
....

....

(1)

13. 4MA1/1FR_Summer_2023_Q12

On the grid, draw the graph of $y = 4 - 2x$ for values of x from -1 to 4



3 marks

14. 4MA1/1FR_Summer_2023_Q5

Here are the first four terms of a number sequence.

2 6 10 14

Elsie correctly works out that the next term in the sequence is 18

(a) Explain how she was able to work this out.

....

(1)

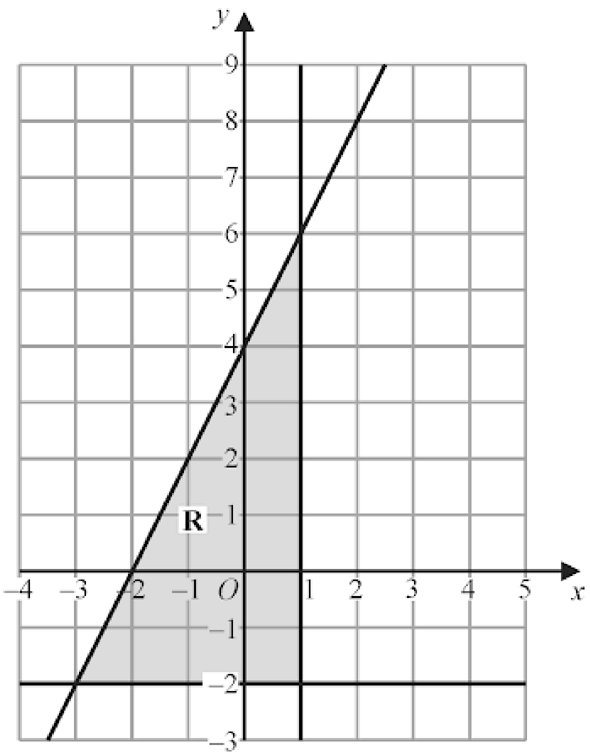
(b) Explain why 217 cannot be a number in the sequence.

....

....

(1)

15. 4MA1/1F_Summer_2023_Q22



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Find the inequalities that define **R**

....

....

....

4 marks

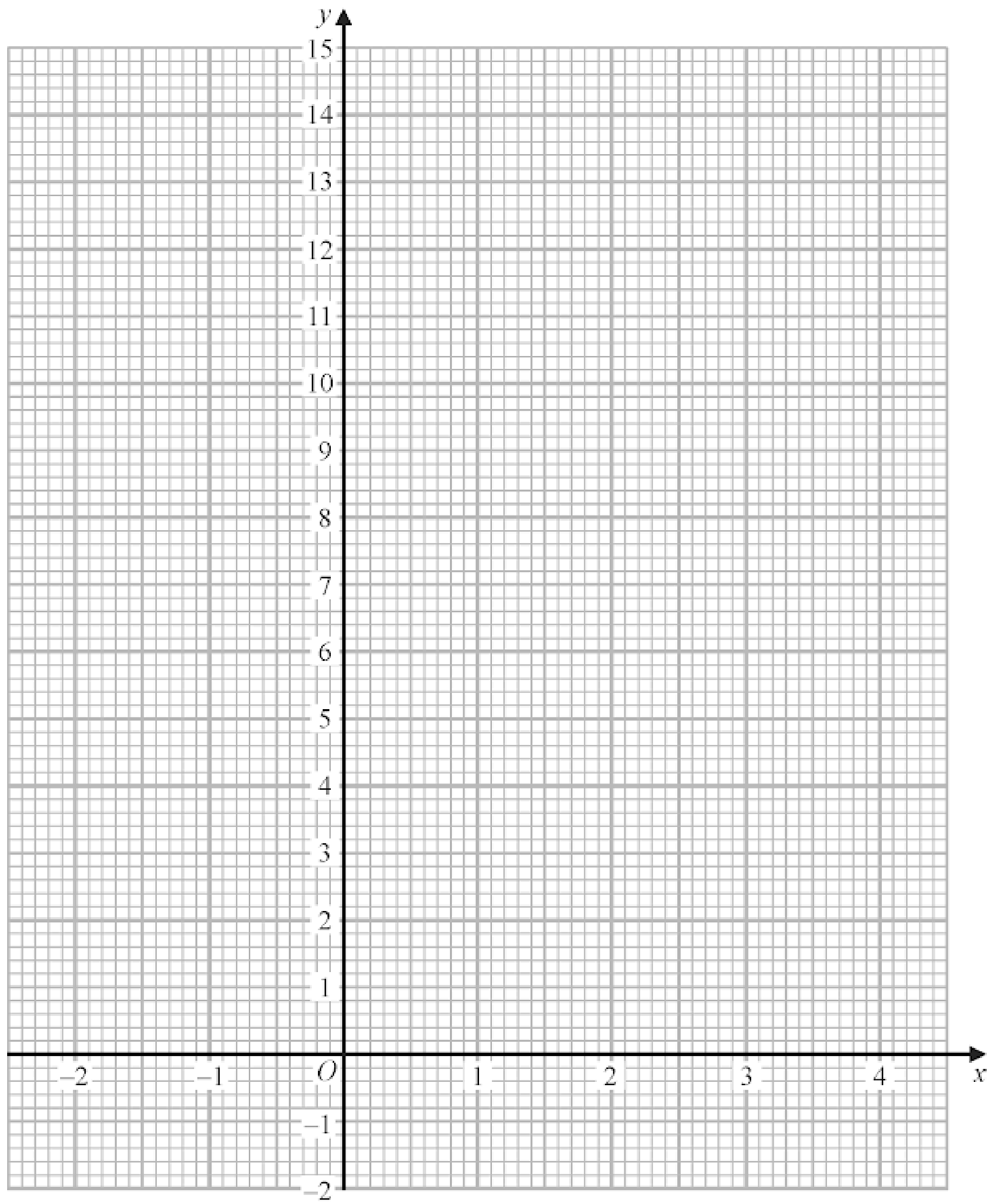
16. 4MA1/1F_Winter_2022_Q18

(a) Complete the table of values for $y = x^2 - 4x + 3$

x	-2	-1	0	1	2	3	4
y		8	3			0	

(2)

(b) On the grid, draw the graph of $y = x^2 - 4x + 3$ for values of x from -2 to 4



(2)

17. 4MA1/1F_Winter_2022_Q5

Here are the first 4 terms of a number sequence.

7 12 17 22

(a) (i) Write down the next term of the sequence.

....

(1)

(ii) Explain how you worked out your answer.

....

(1)

(b) Is 256 a number in the sequence?

Tick one of the boxes below and give a reason for your answer.

Yes ☐ No ☐

Give a reason for your answer.

....

....

(1)

18. 4MA1/1F_Winter_2022_Q25

The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

....

(2)

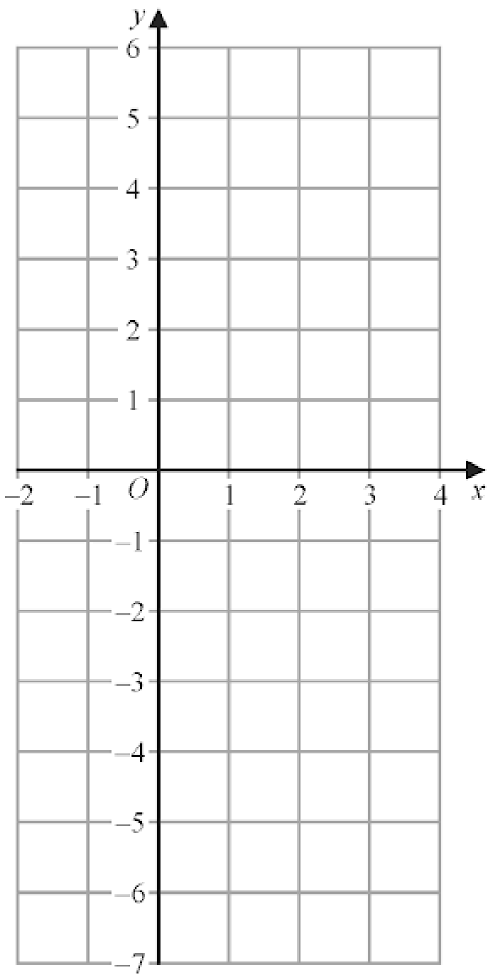
(b) Find the coordinates of the point where **L** crosses the y-axis.

(.................... , ...)

(1)

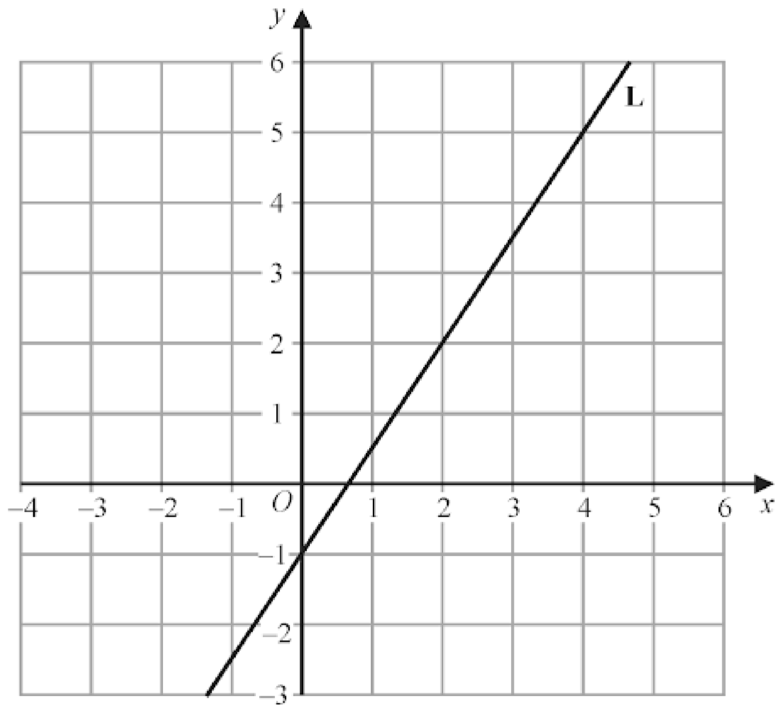
19. 4MA1/1F_Winter_2022_Q14

On the grid, draw the graph of $y = 2x - 3$ for values of x from -2 to 4



3 marks

Line **L** is drawn on the grid.



Find an equation for **L**
Give your answer in the form $y = mx + c$

....

3 marks

21. 4MA1/1FR_Summer_2022_Q18

In a warehouse there are two types of shelves, type **R** and type **S**.

These two types of shelves are arranged into shelving units that form a sequence of patterns.

Here are the first three terms in the sequence.

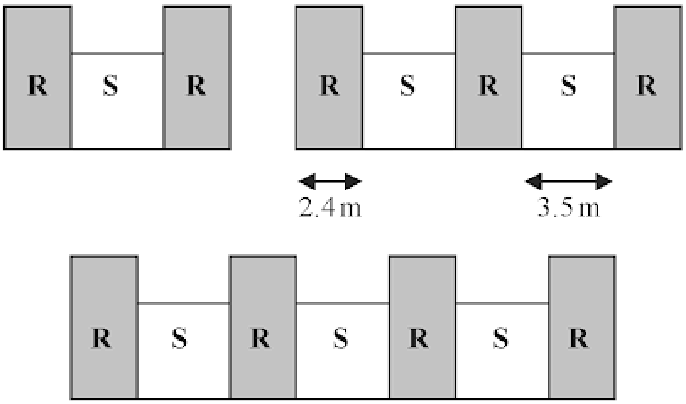


Diagram **NOT** accurately drawn

The width of each type **R** shelf is 2.4 m and the width of each type **S** shelf is 3.5 m

(a) Work out the total width of a shelving unit that has 6 type **R** shelves.

.... m

(2)

A shelving unit has n type **R** shelves.
The total width of this shelving unit is W metres.

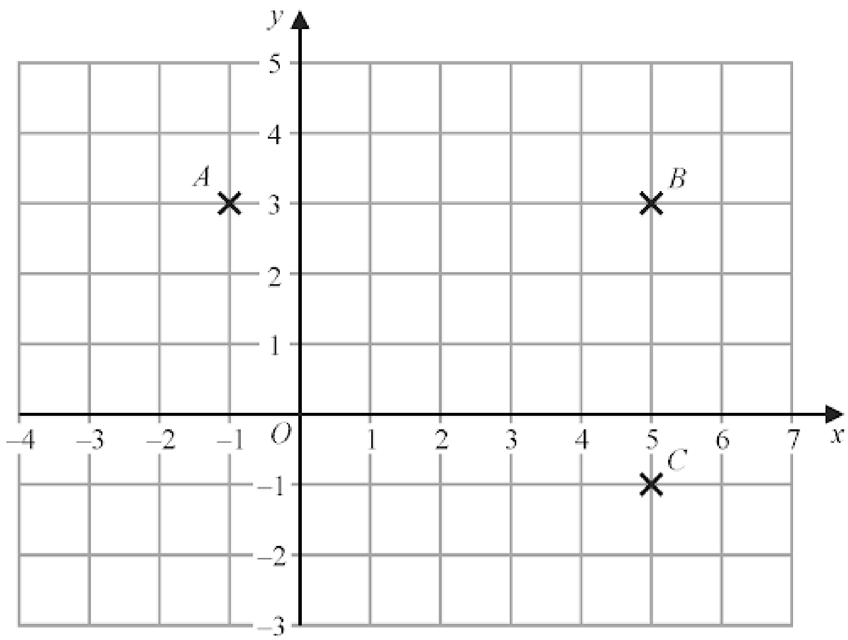
(b) Find an expression for W in terms of n
Give your answer in its simplest form.

$W =$ ....

(2)

22. 4MA1/1F_Summer_2022_Q7

The three points A, B and C are marked on a centimetre grid.



(a) Write down the coordinates of A

(.................... , ...)
(1)

(b) Find the coordinates of the midpoint of BC

(.................... , ...)
(2)

(c) Work out the area of triangle ABC

.... cm^2
(2)

D is the point on the grid so that $ABCD$ is a rectangle.

(d) On the grid, mark with a cross (×) the point D
Label this point D

(1)

23. 4MA1/1F_Summer_2022_Q16

Here are the first five terms of an arithmetic sequence.

$$1 \qquad 5 \qquad 9 \qquad 13 \qquad 17$$

(a) Find an expression, in terms of n, for the nth term of this sequence.

....
(2)

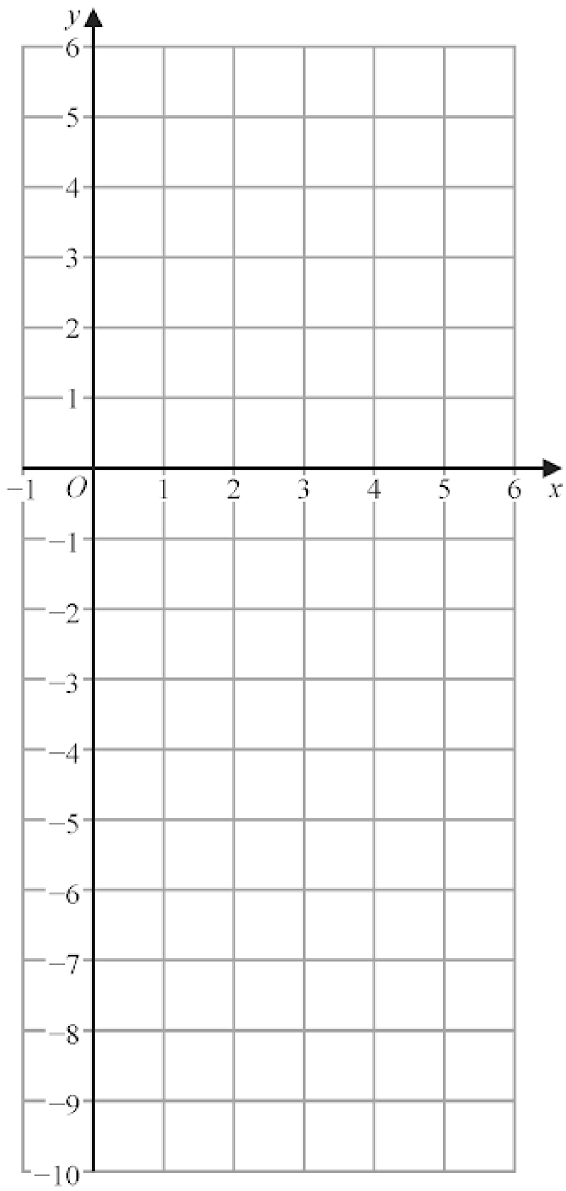
The nth term of another arithmetic sequence is $3n + 5$

(b) Find an expression, in terms of m, for the $(2m)$th term of this sequence.

....
(1)

24. 4MA1/1F_Summer_2022_Q13

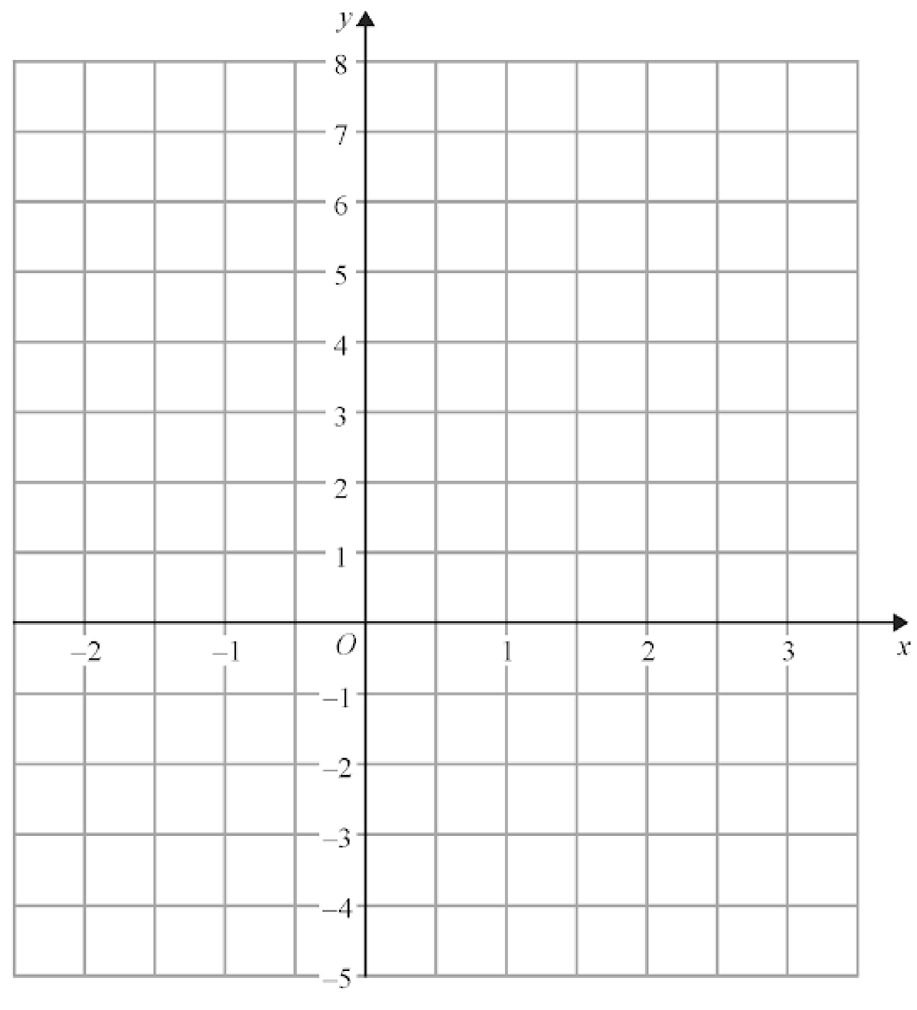
On the grid, draw the graph of $y = -2x + 3$ for values of x from -1 to 5



3 marks

25. 4MA1/1FR_Winter_2021_Q11

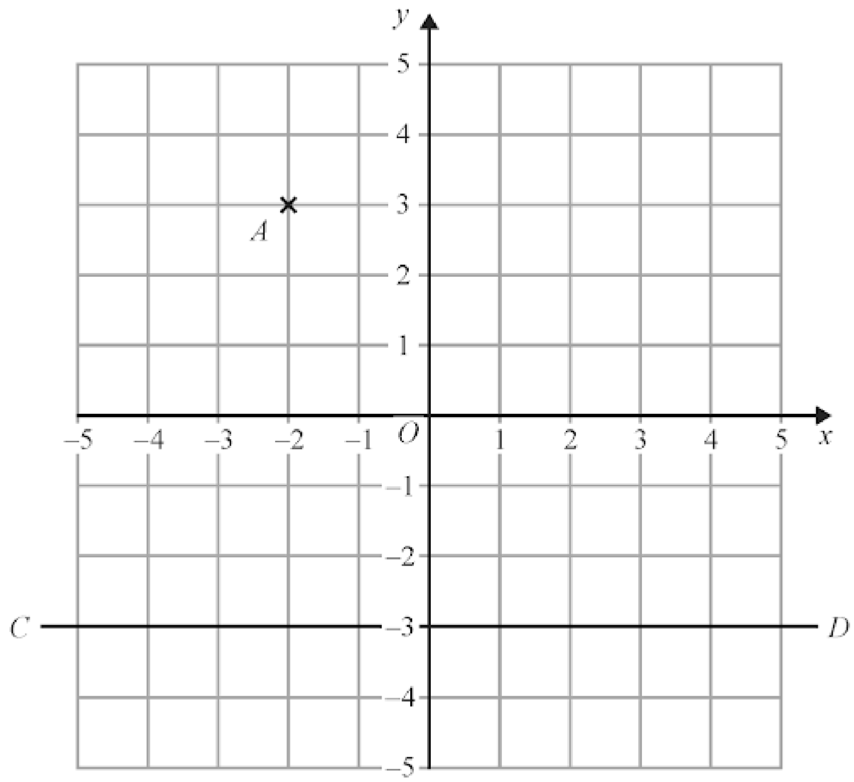
On the grid, draw the graph of $y = 3 - 2x$ for values of x from -2 to 3



3 marks

26. 4MA1/1FR_Winter_2021_Q5

The diagram shows the point A and the line CD on a grid.



(a) Write down the coordinates of point A.

(.................... , ...)

(1)

The point B has coordinates (4, –2)

(b) On the grid, mark with a cross (×) the point B.
Label the point B.

(1)

(c) Write down an equation of the line CD.

....

(1)

27. 4MA1/1F_Summer_2021_Q17

Here are the first 4 terms of an arithmetic sequence.

85 79 73 67

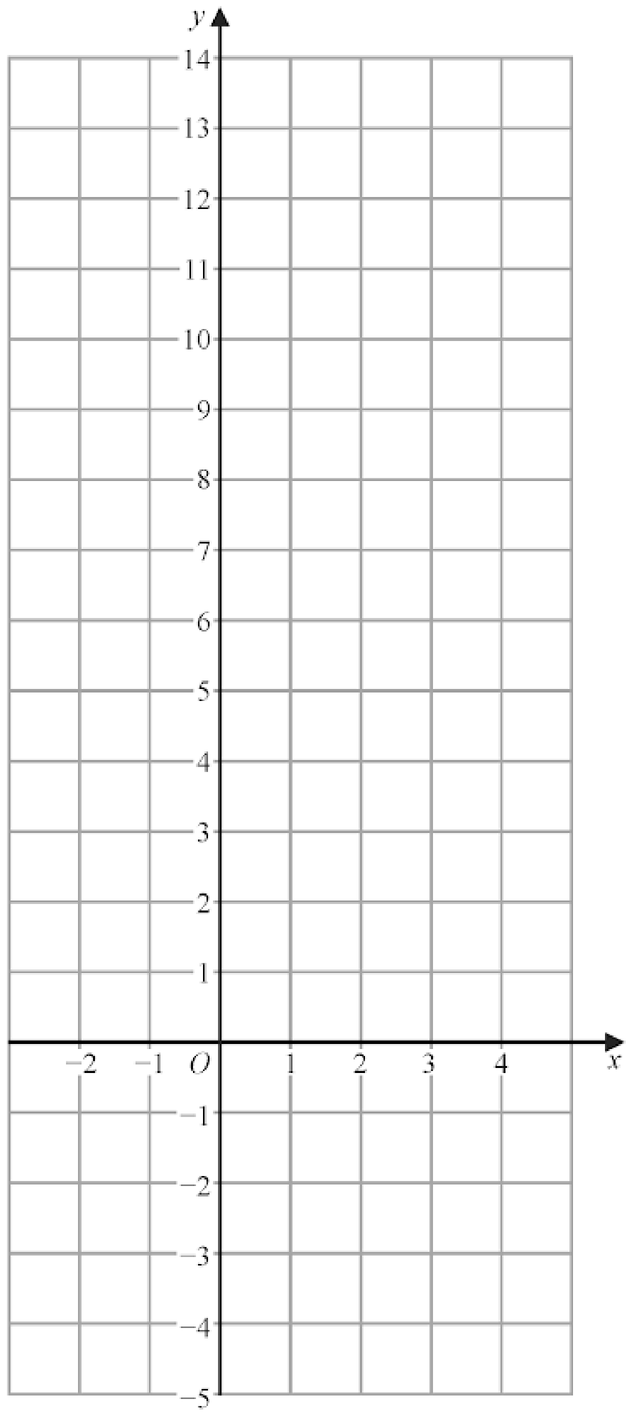
Find an expression, in terms of n, for the nth term of the sequence.

....

2 marks

28. 4MA1/1FR_Winter_2020_Q11

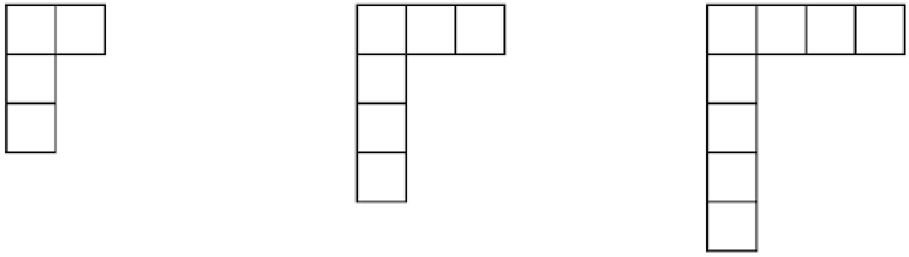
On the grid, draw the graph of $y = 3x + 2$ for values of x from -2 to 4



3 marks

29. 4MA1/1F_Winter_2020_Q4

Here is a sequence of patterns made from square tiles.



Pattern number 1 Pattern number 2 Pattern number 3

(a) In the space below, draw Pattern number 4

(1)

(b) Complete the table.

Pattern number	1	2	3	4	5
Number of tiles	4	6	8		

(1)

(c) Work out the number of tiles in Pattern number 30

....

(2)

Liz says that in Pattern number n, the number of tiles is $2n$.

(d) Is Liz correct?
You must give a reason for your answer.

....

....

(1)

30. 4MA1/1FR_Summer_2020_Q22

31. 4MA1/1F_Summer_2020_Q16

32. 4MA1/1FR_Winter_2019_Q13

33. 4MA1/1FR_Winter_2019_Q9

34. 4MA1/1FR_Winter_2019_Q6

35. 4MA1/1F_Winter_2019_Q12

36. 4MA1/1F_Winter_2019_Q3

37. 4MA1/1FR_Summer_2019_Q11

38. 4MA1/1F_Summer_2019_Q18

39. 4MA1/1FR_Summer_2018_Q22

40. 4MA1/1FR_Summer_2018_Q11

41. 4MA1/1F_Summer_2018_Q19